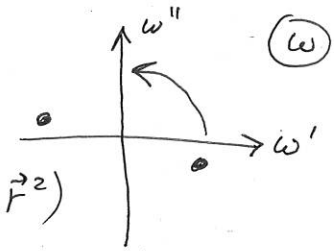
Oct. 20, 2014

Wick rotation and partition function

$$e^{i\hat{H}t/\hbar} \longrightarrow e^{-\hat{H}/k_BT} \quad \text{suggests} \quad \frac{it}{\hbar} \to \beta = \frac{1}{k_BT}$$

$$t = -i\hbar\beta$$

$$\int D\varphi \; e^{\frac{i}{\hbar}\int_0^T dt\, L(\varphi)}$$



Wick rotation: define $\tau = it$, $t = -i\tau$

$$x_\mu x^\mu = c^2t^2 - \vec{r}^{\,2} = -(c^2\tau^2 + \vec{r}^{\,2})$$

$$d^4x = dt\, d\vec{r} = -i\, d\tau\, d\vec{r}$$

Fourier $\varphi(x) = \int \frac{d\omega}{2\pi} \int \frac{d\vec{k}}{(2\pi)^3}\, e^{i(\vec{k}\cdot\vec{r} - \omega t)}\, \varphi(k)$

$$e^{-i\omega t} \to e^{-i\omega(-i\tau)} = e^{-\omega\tau}$$ would blow up for $\omega \to -\infty$

$\Rightarrow$ need to rotate ω as well

$$\omega \to i\omega$$

so $e^{-i\omega t} \to e^{-i(i\omega)(-i\tau)} = e^{-i\omega\tau}$ preserves its form

$$d^4k = d\omega\, d\vec{k} \Rightarrow i\, d\omega\, d\vec{k}$$

Derivatives $\partial_\mu\partial^\mu = \partial^2 = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \vec{\nabla}^2 = -\left(\frac{1}{c^2}\frac{\partial^2}{\partial\tau^2} + \vec{\nabla}^2\right)$

$$\frac{\partial}{\partial t} = \frac{\partial\tau}{\partial t}\frac{\partial}{\partial\tau} = i\frac{\partial}{\partial\tau}$$

$$(\partial_\mu\varphi)^2 = -\left[\frac{1}{c^2}\left(\frac{\partial\varphi}{\partial\tau}\right)^2 + (\vec{\nabla}\varphi)^2\right]$$

so that ~~upto~~ everything shows up with symmetric Euclidean metric $(+,+,+,+)$ instead of Minkowski $(+,-,-,-)$

Action $iS = i\int d^4x \left(\frac{1}{2}(\partial_\mu \varphi)^2 - \frac{1}{2}m^2\varphi^2 - \frac{\lambda}{4!}\varphi^4 + J\varphi\right)$

(Wick rotation) $= i(-i)\int d\tau\, d\vec{r}\left(-\frac{1}{2}\left[(\partial_\tau\varphi)^2 + (\vec{\nabla}\varphi)^2\right] - \frac{1}{2}m^2\varphi^2 - \frac{\lambda}{4!}\varphi^4 + J\varphi\right)$

$= (-)\int d\tau\, d\vec{r}\left\{\frac{1}{2}(\partial_\tau\varphi)^2 + \frac{1}{2}(\vec{\nabla}\varphi)^2 + \frac{1}{2}m^2\varphi^2 + \frac{\lambda}{4!}\varphi^4 - J\varphi\right\}$

$$\int D\varphi\, e^{iS/\hbar} \to \int D\varphi\, e^{-\frac{1}{\hbar}\int d\tau d\vec{r}\{\quad\}}$$

<u>Partition function</u> (for a single particle first)

$$Z = \mathrm{Tr}\left[e^{-\beta\hat{H}}\right] = \sum_m \langle m|e^{-\beta\hat{H}}|m\rangle = \sum_m e^{-\beta E_m}$$

<u>Recall</u> From density matrix $D(x', -i\hbar\beta; x, 0) = \langle x'|e^{-i\hat{H}(-i\beta)}|x\rangle =$

$$= \langle x'|e^{-\beta\hat{H}}\sum_m |m\rangle\langle m|x\rangle =$$

$$= \sum_m e^{-\beta E_m}\langle x'|m\rangle\langle m|x\rangle$$

$$\Rightarrow \int dx\, D(x=x', -i\hbar\beta; x, 0) = \sum_m \int dx\, e^{-\beta E_m}\langle m|x\rangle\langle x|m\rangle =$$

$$= \sum_m e^{-\beta E_m}$$

$$\Rightarrow Z = \int dx\, D(x, -i\hbar\beta; x, 0) = \int dx \int Dx(\tau)\, e^{-\int_0^\beta d\tau \underbrace{\left[\frac{m}{2}\left(\frac{dx}{d\tau}\right)^2 + V(x)\right]}_{\text{Euclidian Lagrangian } L_E(x)}}$$

by repeating "slicing" procedure

What ~~does~~ are boundary conditions?

<u>PBC</u> $x(\tau=0) = x(\tau=\beta)$ because $Z = \mathrm{Tr}\left[e^{-\beta\hat{H}}\right]$

$$\Rightarrow Z = \int_{\text{PBC}} Dx(\tau)\, e^{-\int_0^\beta d\tau\left[\frac{1}{2}m\left(\frac{dx}{d\tau}\right)^2 + V(x)\right]}$$

(periodic BC)

Thus euclidian field theory

$$Z = \mathrm{Tr}\left[e^{-\beta \hat{H}}\right] = \int_{PBC} D\varphi \; e^{-\int_0^\beta d\tau \int d\vec{r}\, L_E(\varphi)}$$

where PBC means $\varphi(0,\vec{r}) = \varphi(\beta,\vec{r})$ (for all $\vec{r}$)

Fourier $\varphi(\tau,\vec{r}) = \sqrt{\frac{\beta}{V}} \sum_{\omega_n} \sum_{\vec{k}} e^{-i\omega_n \tau + i\vec{k}\cdot\vec{r}} \varphi(\omega_n,\vec{k})$

V = spatial volume, $\beta = \frac{1}{k_B T}$ = "temporal" volume

$$\int_0^\beta d\tau \int d^3\vec{r}\, e^{i\omega\tau} e^{-i\vec{k}\cdot\vec{r}} = \beta V \delta_{\vec{k},0}\, \delta_{\omega,0}$$

$$\varphi(0,\vec{r}) = \sqrt{\frac{\beta}{V}} \sum_{\omega_n} \sum_{\vec{k}} e^{i\vec{k}\cdot\vec{r}} \varphi(\omega_n,\vec{k})$$

$$\varphi(\beta,\vec{r}) = \sqrt{\frac{\beta}{V}} \sum_{\omega} \sum_{\vec{k}} e^{-i\omega_n\beta + i\vec{k}\cdot\vec{r}} \varphi(\omega_n,\vec{k})$$

$$\Rightarrow e^{-i\omega_n\beta} = 1 \Rightarrow \omega_n \beta = 2\pi n$$

Matsubara frequency $\omega_n = \frac{2\pi n}{\beta} = 2\pi n k_B T$

Free scalar theory again

$$\mathcal{L}_E = \frac{1}{2}(\partial_\tau \varphi)^2 + \frac{c^2}{2}(\vec{\nabla}\varphi)^2 + \frac{1}{2}m^2\varphi^2$$

Fourier

$$Z = \int D\varphi(\omega_n, \vec{k}) \exp -\frac{1}{2}\beta^2 \sum_{\omega_n, \vec{k}} \varphi(-\omega_n, -\vec{k})\left(\omega_n^2 + c^2\vec{k}^2 + m^2\right)\varphi(\omega_n, \vec{k})$$

$$= \prod_{\omega_n}\prod_{\vec{k}} \int d\varphi(\omega_n, \vec{k}) \exp -\frac{\beta^2}{2}\left(\omega_n^2 + c^2k^2 + m^2\right)|\varphi(\omega_n, \vec{k})|^2$$

$$= \prod_{\omega_n}\prod_{\vec{k}} \sqrt{\frac{2\pi}{\beta^2(\omega_n^2 + c^2k^2 + m^2)}} = \underset{\substack{\uparrow \\ \text{normalization} \\ \text{constant}}}{N} \exp -\frac{1}{2}\sum_{\omega_n}\sum_{\vec{k}} \ln\left[\beta^2(\omega_n^2 + c^2k^2 + m^2)\right]$$

Need $\sum_n \ln(\omega_n^2 + a) = f(a)$ with $a = c^2k^2 + m^2 > 0$

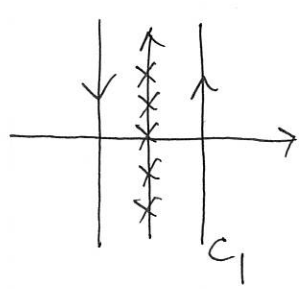
Consider $f'(a) = \frac{d}{da}\sum_n \ln(\omega_n^2 + a) = \sum_n \frac{1}{\omega_n^2 + a}$

This is a HW2 problem!

$$f' = \sum_{n=-\infty}^{\infty} \frac{1}{\left(\frac{2\pi n}{\beta}\right)^2 + a} = \sum_{n=-\infty}^{\infty} \frac{1}{\left(\frac{2}{\beta}\right)^2\left[\pi^2 n^2 + \frac{\beta^2 a}{4}\right]} = \frac{\beta^2}{4}\sum_n \frac{1}{\pi^2 n^2 + z^2}$$

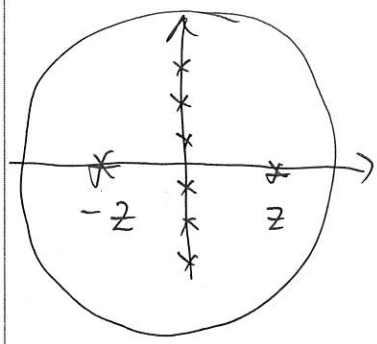
$$= \frac{\beta^2}{4}\int_{C_1} d\omega \, \frac{2}{e^{2\omega} - 1} \, \frac{1}{z^2 - \omega^2}$$

$$z^2 \equiv \frac{1}{4}\beta^2 a$$



Next consider $I = \oint_{C_2} d\omega \, \frac{2}{e^{2\omega}-1} \, \frac{z}{z^2-\omega^2} = 0$

(All singularities inside C_2)



ω

$$I = 2\pi i \sum_{Res} \left(\frac{2}{e^{2\omega}-1} \, \frac{z}{\omega^2 - z^2} \right) =$$

$$= 2\pi i \left(\sum_{\omega = i\pi l} \frac{z}{(i\pi l)^2 - z^2} + \sum_{\omega = \pm z} \right)$$

$$= 2\pi i \left(- \sum_{l} \frac{z}{\pi^2 l^2 + z^2} + \frac{1}{2} \left(\frac{2}{e^{2z}-1} - \frac{2}{e^{-2z}-1} \right) \right)$$

But $\sum_{l} \frac{z}{\pi^2 l^2 + z^2} = \frac{4z}{\beta^2} f'$

$$I = 2\pi i \left\{ - \frac{4z}{\beta^2} f'(a) + \frac{ch(z)}{Sh(z)} \right\} = 0 \qquad \left(z^2 = \frac{1}{4}\beta^2 a \right)$$

$$\Rightarrow \quad f'(a) = \frac{\beta^2}{4z} \frac{ch(z)}{Sh(z)} \Bigg|_{z = \frac{\beta\sqrt{a}}{2}} = \frac{\beta}{2\sqrt{a}} \frac{ch \frac{\beta\sqrt{a}}{2}}{Sh \frac{\beta\sqrt{a}}{2}}$$

We need $f(a) = \int_0^a dx \, f'(x) + f(0)$ $\qquad f'(x) = \frac{1}{2\sqrt{x}} \frac{ch \frac{\beta\sqrt{x}}{2}}{Sh \frac{\beta\sqrt{x}}{2}}$

Let $\frac{\beta\sqrt{x}}{2} = y$, $x = \frac{4}{\beta^2} y^2$, $dx = \frac{8}{\beta^2} y \, dy$

$$f(a) = \int_0^{\frac{\beta}{2}\sqrt{a}} \frac{8}{\beta^2} dy \, y \, \frac{\beta}{2 \cdot \frac{2}{\beta} y} \frac{ch y}{sh y} = \frac{2\beta}{\beta} \int dy \frac{ch y}{Sh y}$$

$$f(a) = \frac{2}{\beta} \int_0^{\frac{\beta}{2}\sqrt{a}} dy \, \frac{\operatorname{ch} y}{\operatorname{sh} y} + f(0) = \frac{2}{\beta} \ln|\operatorname{sh} y| \Big|_0^{\frac{\beta}{2}\sqrt{a}} + f(0)$$

$$= \frac{2}{\beta} \ln\left(\frac{e^{\frac{\beta}{2}\sqrt{a}} - e^{-\frac{\beta}{2}\sqrt{a}}}{2}\right) + \tilde{f}(0)$$

$$= \frac{2}{\beta} \ln e^{\frac{\beta}{2}\sqrt{a}} \left(1 - e^{-\beta\sqrt{a}}\right) + \tilde{\tilde{f}}(0)$$

$$= \beta E_k + 2 \ln\left(1 - e^{-\beta E_k}\right) + \text{const}$$

Because $a = c^2 k^2 + m^2 \equiv E_k^2$, $\sqrt{a} = E_k$.

Thus $Z = N' \exp - \frac{1}{2} \sum_{\vec{k}} \left[\beta E_k + 2 \ln\left(1 - e^{-\beta E_k}\right)\right]$

$$Z = e^{-\beta F} \quad (F = \text{free energy})$$

$$\Rightarrow \quad F = \sum_{\vec{k}} \left\{\frac{1}{2} E_k + \frac{1}{\beta} \ln\left(1 - e^{-\beta E_k}\right)\right\}$$

! recovered ideal gas result from Stat. Mech!

$$\sum_{\vec{k}} \ldots \to V \int \frac{d^3\vec{k}}{(2\pi)^3} \ldots$$